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LINERS HAVING FLOW OPENINGS, AND RELATED CHAMBER KITS, PROCESSING CHAMBERS, AND METHODS FOR SEMICONDUCTOR MANUFACTURING

Abstract

The present disclosure relates to liners having flow openings, and related chamber kits, processing chambers, and methods for semiconductor manufacturing. In one or more embodiments, a liner applicable for semi-conductor manufacturing includes an inner face and an outer face opposing the inner face. The liner includes an inlet opening extending into the inner face and an outlet opening extends into the inner face. The liner includes a curved flow opening extending into the outer face and extending at least partially about the liner. The curved flow opening extends along an azimuthal angle greater than 90 degrees.

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to liners having flow openings, and related chamber kits, processing chambers, and methods for semiconductor manufacturing.

Description of the Related Art

[0002] Semiconductor substrates are processed for a wide variety of applications, including the fabrication of integrated devices and microdevices. One method of processing substrates includes depositing a material, such as a semiconductor material or a conductive material, on an upper surface of the substrate. For example, epitaxy is one deposition process that deposit films of various materials on a surface of a substrate in a processing chamber. During processing, various parameters can affect the uniformity of material deposited on the substrate.

[0003] However, operations (such as epitaxial deposition operations) can involve corrosion, deposition, and/or contamination on chamber components. For example, gases can leak in between surfaces of chamber components, and the gases can condense on surfaces of chamber components. Such leakage can reduce component lifespans, increase gas consumption, and/or contaminate processed substrates.

[0004] Therefore, a need exists for improved apparatuses and methods in semiconductor processing.

SUMMARY

[0005] The present disclosure relates to liners having flow openings, and related chamber kits, processing chambers, and methods for semiconductor manufacturing.

[0006] In one or more embodiments, a liner applicable for semi-conductor manufacturing includes an inner face and an outer face opposing the inner face. An inlet opening extends into the inner face and an outlet opening extends into the inner face. A curved flow opening extends into the outer face and extends at least partially about the liner. The curved flow opening extends along an azimuthal angle greater than 90 degrees.

[0007] In one or more embodiments, a chamber kit applicable for semi-conductor manufacturing includes a first liner with. The first liner includes a first inner face and a first outer face opposing the first inner face. The first liner includes a first flow opening extending into the first outer face and extending at least partially about the first liner. The first flow opening extends along a first azimuthal angle greater than 90 degrees. The chamber kit also includes a second liner. The second liner includes a second inner face and a second outer face opposing the first inner face. The second liner includes a second flow opening sized and shaped to at least partially align with the first flow opening. The second flow opening extends into the second outer face and extends at least partially about the second liner. The second flow opening extends along a second azimuthal angle greater than 90 degrees.

[0008] In one or more embodiments, a processing chamber applicable for semi-conductor manufacturing includes one or more sidewalls at least partially defining a processing volume, a substrate support disposed in the processing volume, and one or more heat sources operable to heat the processing volume. The processing chamber also includes a first liner disposed in the processing volume. The first liner includes a first inner face, a first outer face opposing the first inner face, and a first flow opening that extends into the first outer face and extends at least partially about the first liner. The first flow opening extends along a first azimuthal angle greater than 90 degrees. The processing chamber also includes a second liner at least partially supporting the first liner. The second liner includes a second inner face, a second outer face opposing the first inner face, and a second flow opening at least partially aligning with the first flow opening. The second flow opening extends into the second outer face and extends at least partially about the second liner. The second flow opening extends along a second azimuthal angle greater than 90 degrees.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

[0010] FIG. **1** is a schematic side cross-sectional view of a processing chamber, according to one or more embodiments.

[0011] FIG. **2** is a schematic partial top cross-sectional view of the processing chamber shown in FIG. **1**, according to one or more embodiments.

[0012] FIG. **3** is schematic partial front view of the upper flow module and the lower flow module, shown in FIG. **1** according to one or more embodiments.

[0013] FIG. **4** is a schematic isometric view of a chamber kit applicable for semi-conductor manufacturing.

[0014] FIG. **5** is a schematic isometric view of a chamber kit applicable for semi-conductor manufacturing.

[0015] FIG. **6**A shows a schematic perspective profile view of material buildup on the chamber body using configurations described herein, according to one or more embodiments.

[0016] FIG. **6**B is a schematic perspective profile view of material buildup on the chamber body using other configurations.

[0017] FIG. **7**A is a schematic top isometric view of the upper liner shown in FIGS. **1** and **4**, according to one or more embodiments.

[0018] FIG. **7**B is a schematic bottom isometric view of the upper chamber liner shown in FIG. **7**A, according to one or more embodiments.

[0019] FIG. **7**C is a schematic bottom view of the upper chamber liner shown in FIG. **7**A, according to one or more embodiments.

[0020] FIG. **8**A is a schematic isometric view of the middle chamber liner shown in FIGS. **1** and **4**, according to one or more embodiments.

[0021] FIG. **8**B is a schematic top view of the middle liner shown in FIG. **8**A, according to one or more embodiments.

[0022] FIG. **8**C is a schematic bottom view of the middle liner shown in FIG. **8**A, according to one or more embodiments.

[0023] FIG. **9**A is a schematic isometric view of the lower liner shown in FIGS. **1** and **4**, according to one or more embodiments.

[0024] FIG. **9**B is a schematic top view of the lower liner shown in FIG. **9**A, according to one or more embodiments.

[0025] FIG. **10** is a schematic block diagram view of a method of substrate processing for semiconductor manufacturing, according to one or more embodiments.

[0026] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0027] The present disclosure relates to liners having flow openings, and related chamber kits, processing chambers, and methods for semiconductor manufacturing.

[0028] FIG. **1** is a schematic side cross-sectional view of a processing chamber **100**, according to one or more embodiments. The processing chamber **100** is a deposition chamber. In one or more embodiments, the processing chamber **100** is an epitaxial deposition chamber. The processing chamber **100** is utilized to grow an epitaxial film on a substrate **102**. The processing chamber **100** creates a cross-flow of precursors across a top surface **150** of the substrate **102**. The processing chamber **100** is shown in a processing condition in FIG. **1**.

[0029] The processing chamber **100** includes an upper body **156**, a lower body **148** disposed below the upper body **156**, an upper flow module **113**, and a lower flow module **112** disposed between the upper body **156** and the lower body **148**. The upper body **156**, the upper flow module **113**, the lower flow module **112**, and the lower body **148** form a chamber body. Disposed within the chamber body is a substrate support **106**, an upper window **108** (such as an upper dome), a lower window **110** (such as a lower dome), and one or more heat sources **141**, **143**. The one or more heat sources **141**, **143** include a plurality of upper heat sources **141** and a plurality of lower heat sources **143**. In one or more embodiments, the upper heat sources **141** include upper lamps and the lower heat sources **143** include lower lamps. In one or more embodiments, the lamps include halogen lamps. In one or more embodiments, the lamps are operable to emit infrared light and/or ultraviolet light. The present disclosure contemplates that other heat sources may be used (in addition to or in place of the lamps) for the various heat sources described herein. For example, resistive heaters, light emitting diodes (LEDs), and/or lasers may be used for the various heat sources described herein.

[0030] The substrate support **106** is disposed between the upper window **108** and the lower window **110**. The substrate support **106** supports the substrate **102**. In one or more embodiments, the substrate support **106** includes a susceptor. Other substrate supports (including, for example, a substrate carrier and/or one or more ring segment(s) that support one or more outer regions of the substrate **102**) are contemplated by the present disclosure. The plurality of upper heat sources **141** are disposed between the upper window and a lid **154**. The plurality of upper heat sources **141** form a portion of the upper heat source module **155**.

[0031] The plurality of lower heat sources **143** are disposed between the lower window **110** and a floor **152**. The plurality of lower heat sources **143** form a portion of a lower heat source module **145**. The upper window **108** is an upper dome and/or is formed of an energy transmissive material, such as quartz. The lower window **110** is a lower dome and/or is formed of an energy transmissive material, such as quartz.

[0032] A processing volume **136** and a purge volume **138** are formed between the upper window **108** and the lower window **110**. The processing volume **136** and the purge volume **138** are part of an internal volume defined at least partially by the upper window **108**, the lower window **110**, and one or more liners **163**, **115**, **117**. In one or more embodiments, the chamber includes three liners **163**, **115**, **117**. In one or more embodiments, an upper liner **163**, a middle liner **115**, and a lower liner **117**. The one or more liners **163**, **115**, **117** are disposed inwardly of the chamber body.

[0033] The internal volume has the substrate support **106** disposed therein. The substrate support **106** includes a top surface on which the substrate **102** is disposed. The substrate support **106** is attached to a shaft **118**. In one or more embodiments, the substrate support **106** is connected to the shaft **118** through one or more arms **119** connected to the shaft **118**. The shaft **118** is connected to a motion assembly **121**. The motion assembly **121** includes one or more actuators and/or adjustment devices that provide movement and/or adjustment for the shaft **118** and/or the substrate support **106** within the processing volume **136**.

[0034] The substrate support **106** may include lift pin holes **107** disposed therein. The lift pin holes **107** are each sized to accommodate a lift pin **132** for lifting of the substrate **102** from the substrate support **106** before or after a deposition process is performed. The lift pins **132** may rest on lift pin stops **134** when the substrate support **106** is lowered from a process position to a transfer position. The lift pin stops **134** can include a plurality of arms **139** that attach to a shaft **135**.

[0035] The upper flow module **113** and the lower flow module **112** include one or more gas inlets

114 (e.g., a plurality of gas inlets), one or more purge gas inlets **164** (e.g., a plurality of purge gas inlets), and one or more gas exhaust outlets **116**. The one or more gas inlets **114** and the one or more purge gas inlets **164** are disposed on the opposite side of the flow module **112** from the one or more gas exhaust outlets **116**. The one or more gas exhaust outlets **116** are formed in one or more flow housings **158** (e.g., gas boxes). In one or more embodiments the upper flow module **113** and/or the lower flow module **112** include one or more metallic bodies. The one or more metallic bodies can include, for example, stainless steel and/or aluminum. Other materials are contemplated. In one or more embodiments the upper flow module **113** and lower flow module **112** abut respectively against outer faces of the liners **163**, **115**, **117**. A pre-heat ring **123** is disposed below the one or more gas inlets **114** and the one or more gas exhaust outlets **116**. The pre-heat ring **123** includes a complete ring or one or more ring segments. The pre-heat ring **123** is disposed on top of a support liner **111**, above the one or more purge gas inlets **164**. The one or more liners **163**, **115**, **117** are disposed on an inner surface of the flow module **112** and protects the flow module **112** from reactive gases used during deposition operations and/or cleaning operations. The gas inlet(s) **114** and the purge gas inlet(s) **164** are each positioned to flow a respective one or more process gases P**1** and one or more purge gases P**2** parallel to the top surface **150** of a substrate **102** disposed within the processing volume **136**. The gas inlet(s) **114** are fluidly connected to one or more process gas sources **151** and one or more cleaning gas sources **153**. The purge gas inlet(s) **164** are fluidly connected to one or more purge gas sources **162**. The one or more gas exhaust outlets **116** are fluidly connected to the one or more exhaust pump(s) **157**. The exhaust pump **157** can assist in the controlled deposition of a layer on the substrate **102**. The one or more process gases P**1** supplied using the one or more process gas sources **151** can include one or more reactive gases (such as one or more of silicon (Si), phosphorus (P), and/or germanium (Ge)) and/or one or more carrier gases (such as one or more of nitrogen (N.sub.2) and/or hydrogen (H.sub.2)). The one or more purge gases P**2** supplied using the one or more purge gas sources **162** can include one or more inert gases (such as one or more of argon (Ar), helium (He), and/or nitrogen (N.sub.2)). One or more cleaning gases supplied using the one or more cleaning gas sources **153** can include one or more of hydrogen (H) and/or chlorine (CI). In one or more embodiments, the one or more process gases P**1** include silicon phosphide (SiP) and/or phosphine (PH.sub.3), and the one or more cleaning gases include hydrochloric acid (HCl).

[0036] In one or more embodiments, one or more gases P**3** flow through an upper liner purge inlet **210** into a first curved flow opening **159** of the upper liner **163** and a second curved flow opening **259** of the middle liner **115**. The one or more gases P**3** flow in addition to the flow of the one or more purge gases P**2**. In one or more embodiments, the one or more gases P**3** include one or more purge gases. The one or more gases P**3** can have the same or different composition as the one or more purge gases P**2**. The first curved flow opening **159** and the second curved flow opening **259** are located between the upper flow module **113** and the upper liner **163** and the middle liner **115**. The first and second curved flow openings **159**, **250** together define a flow channel between the upper and lower flow modules **112**, **113** on one side, and the upper liner **163** and the middle liner **115** on another side. The one or more gases P**3** then flow along a curved path in the flow channel around the outside of the upper liner **163** and the middle liner **115** towards the gas exhaust outlet **116**. The one or more gases P**3** flows from the flow channel through one or more through holes **161** located in the middle liner **115**. The one or more gases P**3** then flow through the through holes **161** into the gas exhaust outlet **116** where the one or more gases P**3** can be exhausted through the one or more flow housings **158**.

[0037] The upper flow module **113** and the lower flow module **112** (which can be at least part of a sidewall of the processing chamber **100**) include the one or more gas inlets **114** in fluid communication with the processing volume **136**. The one or more gas inlets **114** are in fluid communication with one or more flow gaps between the upper liner **163** and the middle liner **115**.

[0038] During a deposition operation (e.g., an epitaxial growth operation), the one or more process gases P**1** flow through the one or more gas inlets **114**, through the one or more gaps, and into the processing volume **136** to flow over the substrate **102**.

[0039] The present disclosure contemplates that the one or more purge gases P**2** can be supplied to the purge volume **138** (through the one or more purge gas inlets **164**) during the deposition operation, and exhausted from the purge volume **138**. The one or more purge gases P**2** flow simultaneously with the flowing of the one or more process gases P**1** and/or the one or more gases P**3**. The one or more process gases P**1** are exhausted through gaps between the upper liner **163** and the middle liner **115**. The one or more process gases P**1**, the one or more purge gases P**2**, and the one or more gases P**3** are exhausted through the one or more gas exhaust outlets **116**.

[0040] In one or more embodiments the one or more gases P**3** flow in a second flow channel **261**. The one or more gases P**3** flow from the second flow channel **261** and around the outside of the one or more flow housings **158** through in one or more gaps **187** between the one or more flow housings **158** and the lower flow module **112**. The one or more gases P**3** flow from the one or more gaps **187** and into an outer exhaust volume **188** that is fluidly connected to the exhaust pump **157**. The present disclosure contemplates that that the one or more gases P**3** can be separately exhausted through one or more second gas exhaust outlets that are separate from the one or more gas exhaust outlets **116**.

[0041] During a cleaning operation, one or more cleaning gases flow through the one or more gas inlets **114**, through the one or more gaps (between the upper liner **163** and the middle liner **115**), and into the processing volume **136**.

[0042] The processing system includes one or more sensor devices **195**, **196**, **197**, **198** (e.g., temperature sensors) configured to measure parameter(s) (e.g., temperature(s)) within the processing chamber **100**. In one or more embodiments, the one or more temperature sensor devices **195**, **196**, **197**, **198** include a central sensor device **196** and one or more outer sensor devices **195**, **197**, **198**. A controller **190** (described below) can control the one or more sensor devices **195**, **196**, **197**, **198**, and can conduct method(s) analyzing uniformity of substrate processing using at least one of the one or more sensor devices **195**, **196**, **197**, **198**. In one or more embodiments, the one or more sensor devices **195**, **196**, **197**, **198** each include a sensor that includes one or more of silicon (Si), carbon (C), gallium (Ga), and/or nitrogen (N). In one or more embodiments, the one or more sensor devices **195**, **196**, **197**, **198** each include a silicon sensor, a silicon carbide (SiC) sensor, and/or a gallium nitride (GaN) sensor. In one or more embodiments, each sensor device **195**, **196**, **197**, **198** is a pyrometer and/or optical sensor, such as an optical pyrometer. The present disclosure contemplates that sensor devices other than pyrometers may be used, and/or one or more of the sensor devices **195**, **196**, **197**, **198** can measure properties (such as metrology properties) other than temperature.

[0043] In one or more embodiments, the one or more sensor devices **195**, **196**, **197**, **198** include one or more upper sensor devices **196**, **197**, **198** disposed above the substrate **102** and adjacent the lid **154**, and one or more lower sensor devices **195** disposed below the substrate **102** and adjacent the floor **152**. The present disclosure contemplates that at least one of the one or more lower sensor devices **195** can be vertically aligned below at least one of the upper sensor devices **196**, **196**, **197** (such as outer sensor device **197**).

[0044] Each sensor device **195**, **196**, **197**, **198**, can be a single-wavelength sensor device or a multi-wavelength (such as dual-wavelength) sensor device. In one or more embodiments, the system including the process chamber **100** includes any one, any two, or any three of the four illustrated sensor devices **195**, **196**, **197**, **198**. In one or more embodiments, the process chamber **100** includes one or more additional sensor devices, in addition to the sensor devices **195**, **196**, **197**, **198**. In one or more embodiments, the process chamber **100** may include sensor devices disposed at different locations and/or with different orientations than the illustrated sensor devices **195**, **196**, **197**, **198**.

[0045] As shown, a controller **190** is in communication with the processing chamber **100** and is used to control processes and methods, such as the operations of the methods described herein. The

controller **190** is configured to receive data or input as sensor readings from sensor(s) (such as one or more of the sensor devices **195**, **196**, **197**, **198**). The sensor devices can include, for example: sensor devices that monitor growth of layer(s) on the substrate **102**; and/or sensor devices that monitor temperatures of the substrate **102**, the substrate support **106**, and/or the liners **163**, **115**, **117**. As described the one or more sensor devices can include, for example pyrometers.
[0046] The controller **190** includes a central processing unit (CPU) **193** (e.g., a processor), a memory **191** containing instructions, and support circuits **192** for the CPU **193**. The controller **190** controls various items directly, or via other computers and/or controllers. In one or more embodiments, the controller **190** is communicatively coupled to dedicated controllers, and the controller **190** functions as a central controller.
[0047] The controller **190** is of any form of a general-purpose computer processor that is used in an industrial setting for controlling various substrate processing chambers and equipment, and sub-processors thereon or therein. The memory **191**, or non-transitory computer readable medium, is one or more of a readily available memory such as random access memory (RAM), dynamic random access memory (DRAM), static RAM (SRAM), and synchronous dynamic RAM (SDRAM (e.g., DDR1, DDR2, DDR3, DDR3L, LPDDR3, DDR4, LPDDR4, and the like)), read only memory (ROM), floppy disk, hard disk, flash drive, or any other form of digital storage, local or remote. The support circuits **192** of the controller **190** are coupled to the CPU **193** for supporting the CPU **193**. The support circuits **192** include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. Operational parameters (e.g., a power applied to the heat sources **141**, **143**, a cleaning recipe, and/or a processing recipe) and operations are stored in the memory **191** as a software routine that is executed or invoked to turn the controller **190** into a specific purpose controller to control the operations of the various chambers/modules described herein. The controller **190** is configured to conduct any of the operations described herein. The instructions stored on the memory, when executed, cause one or more of the operations described herein to be conducted in relation to the processing chamber **100**. The controller **190** and the processing chamber **100** are at least part of a system for processing substrates.
[0048] The various operations described herein can be conducted automatically using the controller **190**, or can be conducted automatically or manually with certain operations conducted by a user.
[0049] The controller **190** is configured to control the deposition, the cleaning, the rotational position, the heating, and gas flow through the processing chamber **100** by providing an output to the controls for the sensor devices **195**, **196**, **197**, **198**, the upper heat sources **141**, the lower heat sources **143**, the process gas source **151**, the purge gas source **162**, the motion assembly **121**, and/or the exhaust pump **157**.
[0050] FIG. **2** is a schematic partial top cross-sectional view of the processing chamber **100** shown in FIG. **1**, according to one or more embodiments. For visual clarity purposes, the upper flow module **113** is shown schematically and without hatching in FIG. **2**.
[0051] In one or more embodiments, the purge gas P**2** flows through the upper liner purge inlet **210** and into the first and second curved flow openings **159**, **259**. The first and second curved flow openings **159**, **259** are located on the outer faces of the upper liner **163** and the middle liner **115** (FIG. **1**). The first curved flow opening **159** and the second curved flow opening **259** extend respectively around the upper liner **163** and the middle liner **115** along a respective azimuthal angle A**1**, A**2** greater than 90 degrees. The first and second curved flow openings **159**, **259** define a volume between the upper flow module **113** and the outer faces of the upper liner **163** and the middle liner **115**. The one or more gases P**3** flow along the first and second curved flow openings **159**, **259** from the upper liner purge inlets **210** towards the one or more through holes **161**.
[0052] FIG. **3** is schematic partial front view of the upper flow module **113**, and the lower flow module **112** shown in FIG. **1**, according to one or more embodiments.
[0053] In one or more embodiments, the upper flow module **113** includes a front face **310**. The lower flow module **112** includes a front face **320**. The upper flow module **113** includes a plurality

of upper liner purge inlets **210** disposed in the upper flow module **113** on opposite sides of one another with respect to the front face **310**. The upper liner purge inlets **210** connect to the first curved flow opening **159** (FIG. **2**) that extends between the plurality of upper liner purge inlets **210** along the outer face of the upper liner **163** (FIG. **2**) along the first azimuthal angle A**1** of at least 90 degrees. The lower flow module **112** includes a plurality of lower purge gas inlets **364** disposed in the lower flow module **112** on opposite sides of one another with respect to the front face **320**. The lower purge gas inlets **364** supply the one or more gases P**3** to the second flow channel **261**.

[0054] FIG. **4** is a schematic isometric view of a chamber kit **400** applicable for semiconductor manufacturing, according to one or more embodiments. The chamber kit **400** includes one or more aspects, features components, operations, and/or properties of the processing chamber **100** shown in FIGS. **1-4**.

[0055] The chamber kit **400** includes the upper liner **163**, the middle liner **115**, and the lower liner **117** as shown in FIG. **1**. The liners **163**, **115**, **117** are formed from an opaque material such as silicon carbide (SiC), opaque quartz (e.g., black quartz, white quartz, and/or grey quartz), and/or graphite coated with silicon carbide and/or opaque quartz. The upper liner **163** includes a side flow inlet **414** through which one or more process gases can flow into the process volume **136** along a direction oriented at an angle relative to the one or more process gases P**1** shown in FIG. **1**. The one or more gases P**3** flow through the upper liner purge inlets **210** (FIG. **3**) into the first curved flow opening **159** and the second curved flow opening **259**. The first curved flow opening **159** extends along the first azimuthal angle A**1** greater than 90 degrees around the outer face of the upper liner **163**. The second curved flow opening **259** extends along the second azimuthal angle A**2** greater than 90 degrees along the outer face of the middle liner **115**. The one or more gases P**3** then flow through the one or more through holes **161** (a pair of through holes **161** are shown) extending into the middle liner **115**. The one or more through holes **161** extending between the second curved flow opening **259** and the one or more gas exhaust outlets **116**.

[0056] In one or more embodiments, the one or more gases P**3** flow through the purge gas inlets (FIG. **3**), into the second flow channel **261**. The second flow channel **261** is defined by a third curved flow opening **359** formed in the middle liner **115** and a fourth curved flow opening **459** formed in the lower liner **117**. The third curved flow opening **359** is between the middle liner **115** and the lower flow module **112**, and the fourth curved flow opening **459** is between the lower liner **117** and the lower flow module **112** to define the second flow channel **261**. The third curved flow opening **359** extends along a third azimuthal angle A**3** greater than 90 degrees around the outer face of the middle liner **115**. The fourth curved flow opening **459** extends along a fourth azimuthal angle A**4** greater than 90 degrees along the outer face of the lower liner **117**. The third curved flow opening **359** is a recess in the outer face of the middle liner **115** and the fourth curved flow opening **459** is a recess in the outer face of the lower liner **117**. The one or more gases P**3** flow along the third and fourth curved flow openings **359**, **459** around the flow housings **158** through the one or more gaps **187**.

[0057] In one or more embodiments, one or more (such as one or all) of the azimuthal angles A**1**-A**4** of the first, second, third, and curved flow openings are the same. In one or more embodiments, the first azimuthal angle A**1** is equal to the second azimuthal angle A**1**. In one or more embodiments the one or more (such as one or all) of the azimuthal angles A**1**-A**4** of the first, second, third, and fourth flow openings are different from each other. In one or more embodiments, the third angle A**3** and/or the fourth angle A**4** are less than the first angle A**1** and/or the second angle A**2**. In one or more embodiments one or more (such as one or all) of the azimuthal angles A**1**-A**4** of the first, second, third, and fourth flow openings are greater than 180 degrees, such as greater than 270 degrees, for example about 360 degrees. In one or more embodiments, the first azimuthal angle A**1** and/or the second azimuthal angle A**2** is greater than 270 degrees, for example about 360 degrees.

[0058] FIG. **5** is a schematic isometric view of a chamber kit **500** applicable for semiconductor manufacturing, according to one or more embodiments. The chamber kit **500** includes one or more aspects, features components, operations, and/or properties of the chamber kit **400** in FIG. **4**.

[0059] In one or more embodiments, the one or more gases P**3** flow through the upper liner purge inlets **210** (FIG. **3**) into a first curved flow opening **559**. The first curved flow opening **559** extends along a first azimuthal angle AA**1** greater than 90 degrees around the outer face of an upper liner **563**. A second curved flow opening **561** extends along a second azimuthal angle AA**2** greater than 90 degrees around the outer face of a middle liner **515**. The one or more gases P**3** flow along the first and second curved flow openings **559**, **561** and into the one or more flow housings **158** (FIG. **1**). In one or more embodiments, the first azimuthal angle AA**1** and the second azimuthal angle AA**2** are respectively less than 330 degrees, such as less than 300 degrees. In one or more embodiments, the first azimuthal angle AA**1** and the second azimuthal angle AA**2** are respectively greater than 180 degrees.

[0060] In one or more embodiments the one or more gases P**3** flow through the lower purge gas inlets **364** (FIG. **3**) and into a third curved flow opening **562** and a fourth curved flow opening **564**. The third curved flow opening **562** is a recess in the outer face of the middle liner **515** and the fourth curved flow opening **564** is a recess in the outer face of a lower liner **517**. The one or more gases P**3** flow along the third and fourth curved flow openings **562**, **564** around the one or flow housings **158** and to the outer exhaust volume **188** through the gaps **187** (FIG. **1**). The third curved flow opening **562** extends along a third azimuthal angle AA**3** less than 180 degrees (such as less than 90 degrees) around the outer face of the middle liner **515**. The fourth curved flow opening **564** extends along a fourth azimuthal angle AA**4** less than 180 degrees (such as less than 90 degrees) around the outer face of the lower liner **517**.

[0061] The present disclosure contemplates that the middle liner **515** and the lower liner **517** can respectively include an additional third curved flow opening **562** and an additional fourth curved flow opening **564** on opposing sides (e.g., on a back side of the view shown in FIG. **5**) of the middle liner **515** and the lower liner **517**.

[0062] FIG. **6**A is a schematic perspective profile view of material buildup on the chamber body using configurations described herein, according to one or more embodiments. The material buildup is on, for example, inner surfaces of the lower flow module **112** and the upper flow module **113**. In one or more embodiments, the material buildup includes chlorine.

[0063] The profile view includes a first profile **601** having a first material concentration, a second profile **602** having a second material concentration higher than the first material concentration, and a third profile **603** having a third material concentration higher than the third material concentration.

[0064] FIG. **6**B is a schematic perspective profile view of material buildup on the chamber body using other configurations.

[0065] As shown by comparing FIG. **6**A with FIG. **6**B, the subject matter described herein facilitates reduced buildup on the chamber body, as shown at least by the smaller third profile **603** in FIG. **6**A compared to the third profile **603** in FIG. **6**B. The reduced buildup can be facilitated, for example, at and/or near an inject area and an exhaust area of the chamber body.

[0066] FIG. **7**A is a schematic top isometric view of the upper liner **163** shown in FIGS. **1** and **4**, according to one or more embodiments.

[0067] The upper liner **163** includes an inner face **701** and the outer face **702** opposing the inner face **701**. The upper liner **163** includes an inlet opening **703** extending into the inner face **701** and an outlet opening **704** extending into the inner face **701**. The first curved flow opening **159** extends into the outer face **702**. The upper liner **163** includes a first side face **715** (e.g., a top face) and a second side face **712** (e.g., a bottom face) opposing the first side face **715**. The first side face **715** and the second side face **712** extending between the inner face **701** and the outer face **702**. In one or more embodiments, the first curved flow opening **159** is a recess extending into a second side face **712** of the upper liner **163** to define a shoulder **711** relative to the second side face **712**. The upper liner **163** includes a flow protrusion **710** which is located on the outer face of the upper liner **163**. The flow protrusion **710** includes a protrusion that extends into the first curved flow opening **159** which extends around the outer face **702** of the upper liner along the first azimuthal angle A**1**. In one or more embodiments a cross-flow gas can be injected through the flow protrusion **710** and flow across the processing volume **136** (FIG. **4**) about perpendicular to the process gas P**1** (FIG. **1**). The flow protrusion **710** sized and shaped to extend into the first curved flow opening **159** such that the one or more gases P**3** flowing in the first curved flow opening **159** can flow around (e.g., above and/or below) the flow protrusion **710**. The one or more gases P**3** flow from the upper liner purge inlets **210** (FIG. **2**), through the first curved flow opening **159**, and around the flow protrusion **710** and toward the **161** (FIG. **4**).

[0068] FIG. **7**B is a schematic bottom isometric view of the upper liner **163** shown in FIG. **7**A, according to one or more embodiments.

[0069] FIG. **7**C is a schematic bottom view of the upper liner **163** shown in FIG. **7**A, according to one or more embodiments.

[0070] FIG. **8**A is a schematic isometric view of the middle liner **115** shown in FIGS. **1** and **4**, according to one or more embodiments.

[0071] The middle liner **115** includes an inner face **801**, an outer face **802**, an inlet opening **803**, and one or more exhaust opening(s) **804** (a plurality is shown). The second curved flow opening **259** extends into the outer face **802**. In one or more embodiments, the second curved flow opening **259** is a recess extending into a first side face **812** (e.g., a top face) of the middle liner **115** to define a shoulder **814** relative to the first side face **812**. The middle liner **115** includes a second side face **815** (e.g., a bottom face) opposing the first side face **812**. The first side face **812** and the second side face **815** extending between the inner face **801** and the outer face **802**. The middle liner **115** includes a protrusion **810** which is located on the outer face **802** of the middle liner **115**. The protrusion **810** extends into the second curved flow opening **259** which extends around the outer face **802** of the middle liner **115** along the second azimuthal angle A**2**. In one or more embodiments the cross-flow gas can be injected through the protrusion **810** and flow across the processing volume **136** (FIG. **4**) about perpendicular to the process gas P**1** (FIG. **1**). The protrusion **810** is sized and shaped to extend into the second curved flow opening **259** such that the one or more gases P**3** flowing in the second curved flow opening **259** can flow around (e.g., above and/or below) the protrusion **810**. The one or more gases P**3** flow from the upper liner purge inlets **210** (FIG. **2**) within the second curved flow opening **159** and around the protrusion **810**, through the through holes **161** and to the exhaust opening **804** at the opposite end of the middle liner **115**. The middle liner **115** includes one more lower flow openings **811** through which the one or more purge gases P**2** can flow though the exhaust opening(s) **804** to the one or more gas exhaust outlets **116** (FIG. **1**). The middle liner **115** includes the third curved flow opening **359** which extends around the outer face **802** along the third azimuthal angle A**3**. In one or more embodiments, the one or more through holes **161** interface with the one or more exhaust openings **804** formed in the middle liner **115**. The one or more exhaust openings **804** are at least part of the one or more gas exhaust outlets **116** shown in FIG. **1**. The one or more through holes **161** extend into a recessed outer surface **813** at least partially defined by the second curved flow opening **259** of the middle liner **115**.

[0072] FIG. **8**B is a schematic top view of the middle liner **115** shown in FIG. **8**A, according to one or more embodiments.

[0073] FIG. **8**C is a schematic bottom view of the middle liner **115** shown in FIG. **8**A, according to one or more embodiments.

[0074] FIG. **9**A is a schematic isometric view of the lower liner **117** shown in FIGS. **1** and **4**, according to one or more embodiments.

[0075] The lower liner **117** includes an inner face **901**, an outer face **902**, an inlet opening **903**, one or more exhaust openings **904** (a plurality is shown), one or more ridges **905** (a plurality is shown),

and the fourth curved flow opening **459** which extends around the outer face **902** along the fourth azimuthal angle A**4**.
[0076] FIG. **9**B is a schematic top view of the lower liner **117** shown in FIG. **9**A, according to one or more embodiments.
[0077] The present disclosure contemplates that the upper liner **163** can be referred to as a first liner, the middle liner **115** can be referred to as a second liner, and/or the lower liner **117** can be referred to as a third liner.
[0078] FIG. **10** is a schematic block diagram view of a method **1000** of substrate processing for semiconductor manufacturing, according to one or more embodiments.
[0079] Operation **1001** includes positioning a substrate on a substrate support in a processing volume of a processing chamber. In one or more embodiments, the positioning includes moving a substrate support and/or a plurality of lift pins relative to each other to land the substrate on the substrate support.
[0080] Operation **1002** of the method **1000** includes heating the substrate to a target temperature.
[0081] Operation **1003** includes flowing one or more process gases over the substrate.
[0082] Operation **1004** includes flowing one or more purge gases. In one or more embodiments, at least part of the purge gases flow around a chamber liner in a curved flow opening to a gas outlet. In one or more embodiments, a pressure is maintained inside the curved flow opening. In one or more embodiments operation **1004** is performed at least partially simultaneously with operation **1103**.
[0083] Benefits of the present disclosure include reduced or eliminated chamber component corrosion, chamber component deposition, and/or chamber substrate contamination; reduced leakage of processing gases; and reduced or eliminated condensation of process gases on chamber components; reduced gas consumption and gas waste; and increased chamber component lifespans; increased growth rates; and more uniform film growth and/or dopant concentration. The chamber components can include, for example, flow modules (e.g., stainless steel or aluminum flow modules) and/or liners (e.g., transparent or opaque liners).
[0084] It is contemplated that one or more aspects disclosed herein may be combined. As an example, one or more aspects, features, components, operations and/or properties of the processing chamber **100**; the upper liner **163**; the middle liner **115**; the lower liner **117**; the upper flow module **113**; the lower flow module **112**; the upper liner purge inlet **210**; the lower purge gas inlets **364**, the gas inlet **114**; the purge gas inlet **164**; the gas exhaust outlet **116**; the first curved flow opening **159**; the second curved flow opening **259**, the through hole(s) **161**; the flow housings **158**; the gaps **187**; the outer exhaust volume **188**; the third curved flow opening **359**; the fourth curved flow opening **459**; the upper liner **563**; the middle liner **515**; the lower liner **517**; the flow protrusion **710**; the protrusion **810**; and/or the method **1000** may be combined. Moreover, it is contemplated that one or more aspects disclosed herein may include some or all of the aforementioned benefits.
[0085] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A substrate processing chamber, comprising: one or more sidewalls at least partially defining a processing volume, the one or more sidewalls comprising a metallic body; a substrate support disposed in the processing volume; one or more heat sources operable to heat the processing volume; a first liner disposed in the processing volume, the first liner disposed inwardly of the metallic body and abutting against the metallic body, the first liner comprising: a first inner face; a first outer face opposing the first inner face; and a first flow opening extending into the first outer face and extending at least partially about the first liner, the first flow opening extending along a first azimuthal angle greater than 90 degrees, the first flow opening defining a flow channel between the first liner and the metallic body.

2. The processing chamber of claim 1, wherein the processing chamber further comprises: a second liner at least partially supporting the first liner, the second liner comprising: a second inner face; a second outer face opposing the first inner face; a second flow opening at least partially aligning with the first flow opening, the second flow opening extending into the second outer face and extending at least partially about the second liner, the second flow opening extending along a second azimuthal angle greater than 90 degrees; and a third flow opening extending into the second outer face and extending at least partially about the second liner, the third flow opening extending along a third azimuthal angle greater than 90 degrees.

3. The processing chamber of claim 2, wherein the third azimuthal angle is greater than 180 degrees.

4. The processing chamber of claim 2, wherein the processing chamber further comprises a third liner at least partially supporting the second liner, the third liner comprising: a third inner face; a third outer face opposing the first inner face; and a fourth flow opening at least partially aligning with the third flow opening, the fourth flow opening extending into the third outer face and extending at least partially about the third liner, the fourth flow opening extending along a fourth azimuthal angle greater than 90 degrees.

5. The processing chamber of claim 2, wherein the first flow opening is a recess extending into the first outer face, and the processing chamber further comprises: a ring disposed between the first liner and the second liner.

6. The processing chamber of claim 2, wherein the processing chamber further comprises: a gas inlet; and a gas exhaust.

7. The processing chamber of claim 6, wherein the second liner further comprises: one or more through holes extending between the second inner face and the second outer face.

8. The processing chamber of claim 7, wherein the one or more through holes interface with one or more exhaust openings formed in the second liner.

9. A liner for a substrate processing chamber, comprising: an inner face; an outer face opposing the inner face; an inlet opening extending into the inner face; an outlet opening extending into the inner face; and a curved flow opening extending into the outer face and extending at least partially about the liner, the curved flow opening extending along an azimuthal angle greater than 90 degrees.

10. The liner of claim 9, wherein the liner comprises an opaque material.

11. The liner of claim 9, wherein the azimuthal angle is greater than 180 degrees.

12. The liner of claim 9, wherein the curved flow opening is a recess that at least partially defines a shoulder along the outer face.

13. The liner of claim 9, wherein the liner further comprises: a first side face extending between the inner face and the outer face; and a second side face extending between the inner face and the outer face, the second side face opposing the first side face, wherein the curved flow opening, the inlet opening, and the outlet opening extend into the second side face.

14. A chamber kit for a substrate processing chamber, comprising: a first liner comprising: a first inner face; a first outer face opposing the first inner face; and a first flow opening extending into the first outer face and extending at least partially about the first liner, the first flow opening extending along a first azimuthal angle greater than 90 degrees; and a second liner comprising: a second inner face; a second outer face opposing the first inner face; and a second flow opening sized and shaped to at least partially align with the first flow opening, the second flow opening extending into the second outer face and extending at least partially about the second liner, the second flow opening extending along a second azimuthal angle greater than 90 degrees.

15. The chamber kit of claim 14, wherein the second flow opening is a recess.

16. The chamber kit of claim 14, wherein the second liner further comprises: a third flow opening extending into the second outer face and extending at least partially about the second liner, the third

flow opening extending along a third azimuthal angle greater than 90 degrees.
17. The chamber kit of claim 16, wherein the second liner further comprises: a first side face extending between the second inner face and the second outer face; and a second side face extending between the second inner face and the second outer face, the second side face opposing the first side face, wherein the second flow opening extends into the first side face and the third flow opening extends into the second side face.
18. The chamber kit of claim 14, wherein the second liner further comprises: one or more through holes extending between the second inner face and the second outer face.
19. The chamber kit of claim 18, wherein the one or more through holes extend into a recessed outer surface at least partially defined by the second flow opening of the second liner.
20. The chamber kit of claim 14, wherein the first azimuthal angle and the second azimuthal angle are respectively greater than 180 degrees.